

Bernoulli Distribution

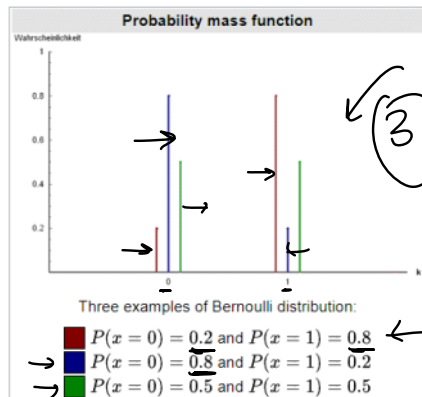
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$p \rightarrow$ prob of success $\rightarrow \hat{p}$
 $1-p \rightarrow$ prob of failure
Experiment

Bernoulli distribution is a probability distribution that models a binary outcome, where the outcome can be either success (represented by the value 1) or failure (represented by the value 0). The Bernoulli distribution is named after the Swiss mathematician Jacob Bernoulli, who first introduced it in the late 1600s.

The Bernoulli distribution is characterized by a single parameter, which is the probability of success, denoted by p . The probability mass function (PMF) of the Bernoulli distribution is:

$$P(X=x) = p^x (1-p)^{1-x}$$



$$P(X=1) = \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^0 = \frac{1}{2}$$

rolling a dice
getting a 5

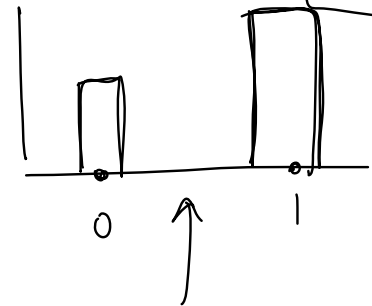
$$\left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^{\sim 5}$$

coin toss $\rightarrow$

binary outcome

$$P(X=1) = \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^0 = \frac{1}{2}$$

$$P(X=0) = \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{1-0} = \frac{1}{2}$$



machine learning

The Bernoulli distribution is commonly used in machine learning for modelling binary outcomes, such as whether a customer will make a purchase or not, whether an email is spam or not, or whether a patient will have a certain disease or not.

Binomial distribution

Binomial Distribution

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(p)

(n, p)

$n=1 \rightarrow$ binomial
Bernoulli

Bernoulli trial

n times

5

$\rightarrow$ Binomial

Binomial distribution is a probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials with two possible outcomes (often called "success" and "failure"), where the probability of success is constant for each trial. The binomial distribution is characterized by two parameters: the number of trials n and the probability of success p .

Feedback

$\rightarrow$ 10 students

0.5

The Probability of anyone watching this lecture in the future and then liking it is 0.5. What is the probability that:

1. No-one out of 3 people will like it

1. 1 out of 3 people will like it

1. 2 out of 3 people will like it

1. 3 out of 3 people will like it

Binomial

$$P(X=x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

$n \rightarrow$ # of trials

$p \rightarrow$ prob of success

$x \rightarrow$ desired result.

PDF Formula:

$$\frac{3!}{2!1!} \times \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1$$

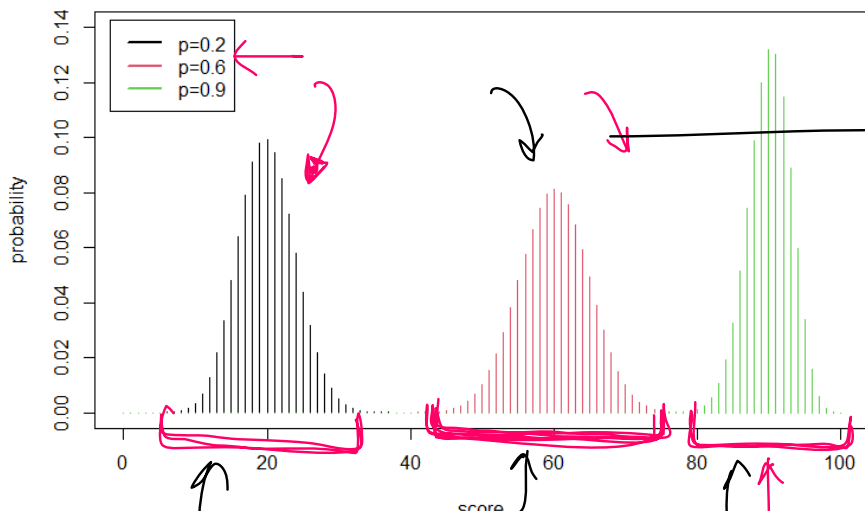
$$3 \times \frac{1}{8} = \frac{3}{8}$$

Graph of PDF:

H T H T H T _ _ 10

H _ _ _ _ _

Binomial distribution with different probabilities of success



$p(s)$ increase

Criteria:

Criteria:

1. The process consists of n trials
2. Only 2 exclusive outcomes are possible, a success and a failure.
3. $P(\text{success}) = p$ and $P(\text{failure}) = 1-p$ and it is fixed from trial to trial
4. The trials are independent.

1. **Binary classification problems:** In binary classification problems, we often model the probability of an event happening as a binomial distribution. For example, in a spam detection system, we may model the probability of an email being spam or not spam using a binomial distribution.
2. **Hypothesis testing:** In statistical hypothesis testing, we use the binomial distribution to calculate the probability of observing a certain number of successes in a given number of trials, assuming a null hypothesis is true. This can be used to make decisions about whether a certain hypothesis is supported by the data or not.
3. **Logistic regression:** Logistic regression is a popular machine learning algorithm used for classification problems. It models the probability of an event happening as a logistic function of the input variables. Since the logistic function can be viewed as a transformation of a linear combination of inputs, the output of logistic regression can be thought of as a binomial distribution.
4. **A/B testing:** A/B testing is a common technique used to compare two different versions of a product, web page, or marketing campaign. In A/B testing, we randomly assign individuals to one of two groups and compare the outcomes of interest between the groups. Since the outcomes are often binary (e.g., click-through rate or conversion rate), the binomial distribution can be used to model the distribution of outcomes and test for differences between the groups.